

Topic 3: Some Basic Operations on Images

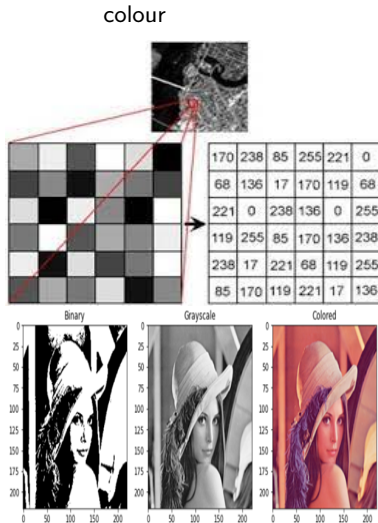
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- 1 Types of Images
- 2 Types of Low Level Image Processing
 - Spatial Domain Processing
 - Point Processing
 - Neighbourhood Processing
 - Transformed Domain Processing
 - Arithmetic operation, considering image as function
- 3 Acknowledgements

- ▶ Binary Image: Binary Matrix
- ▶ Gray scale Image: Matrix with integers from 0 to (L-1)
- ▶ Colour Image: (M_R, M_G, M_B)
 - M_R is matrix with integers from 0 to (L-1), representing intensity of red colour
 - M_G is matrix with integers from 0 to (L-1), representing intensity of green colour
 - M_B is matrix with integers from 0 to (L-1), representing intensity of blue colour

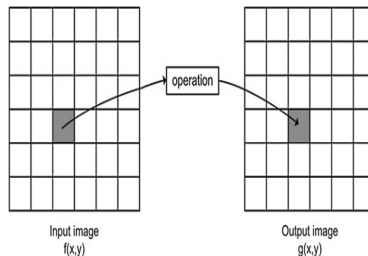


Types of Low Level Image Processing



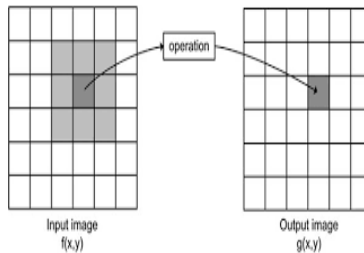
► Spatial Domain

- Point Processing
- Neighbourhood Processing



► Transformed Domain

- Fourier Domain
- Wavelet Domain
- SVD Domain

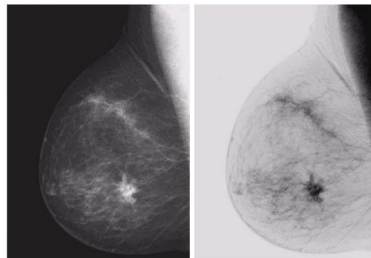
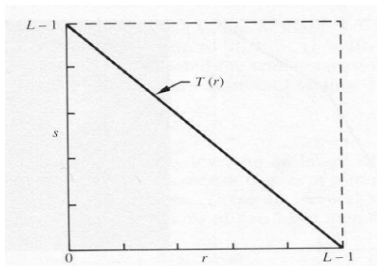


► Defn of Point Processing:

- Let r be the pixel value in input image f
- The output pixel value $s = T(r)$
- T is called as point processing transformation

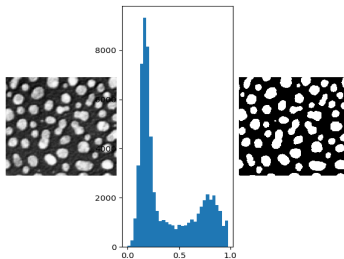
► Examples of point processing

- $s = \text{Max} - r$, where Max is maximum allowed pixel value in the image
- Let $f(x,y)$ be input image
- The output image $g(x,y) = \text{Max} - f(x,y)$

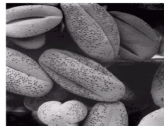
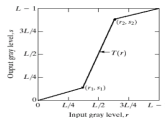


- Image Thresholding:

- $s=0$ if $r \leq T_0$;
 $s=1$ otherwise



- In Other words,
 $g(x,y)=0$ if
 $f(x,y) \leq T_0$
 $g(x,y)=1$ otherwise



- Contrast Stretching

► Definition of Neighbourhood Processing:

To transform the pixel value at a location, consult the pixel values of the neighbours also

► Motivation:

- The neighbours of pixel at (x, y) will provide information about the pixel value at (x, y) . This information can be used

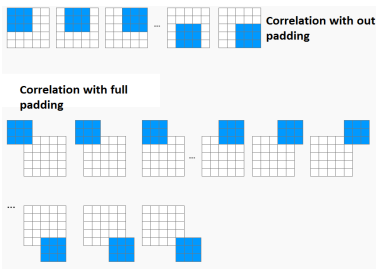
► Examples of Neighbourhood Processing

- Mean filter
- Median Filter



► Correlation

- Correlation without padding
- Correlation with padding
 - Padding is done such a way that the size of the output is same as that of input
 - padding is done infinitely(Full Correlation)
- Application: Template Matching using Normalized cross correlation



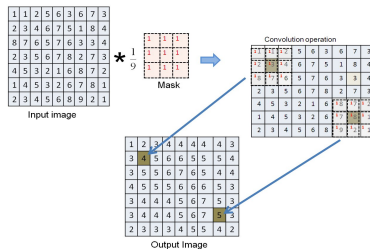
Neighbourhood Processing (cont.)



- Convolution:

$$\text{Conv}(f, w) = \text{cor}(f, w_{180})$$

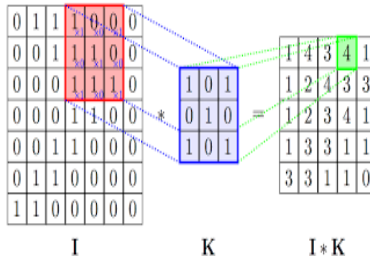
- Full Convolution (Padding is done infinitely)
- Partial Convolution



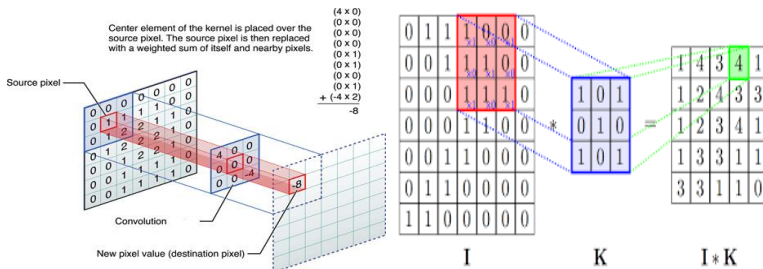
- Padding is done such a way that the size of the output is same as that of input

- No padding

- Application: Polynomial Multiplication



Partial Convolution: Padding such that size of input and output images are the same





Convolution Vs Correlation

► $\text{Con}(f, w) = \text{Cor}(f, w_{180})$

Questions

- Find the size of the $\text{cov}(f, w)$, where size of f is $m \times n$ and the size of w is $a \times b$ when
 - Conv is full (infinite padding)
 - No padding
- Give an $O(n \log n)$ algorithm to find product of two polynomials of degree n (Hint: use Conv)
- Find convolution and correlation 1) with full padding, 2) with partial padding, 3) without padding for the following inputs

Neighbourhood Processing (cont.)



3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6 X 6 image



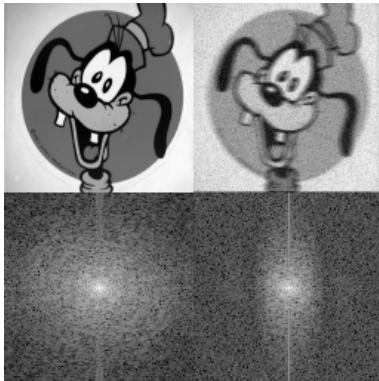
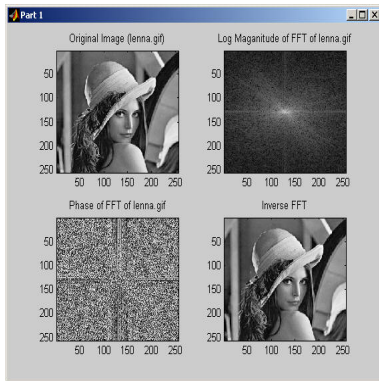
1	0	-1
1	0	-1
1	0	-1

3 X 3 filter



- ▶ Fourier Transform Domain Processing
- ▶ Let $f(x, y)$ be an image, the Discrete Fourier Transform(DFT) of $f(x, y)$ is denoted as $F(u, v)$, and defined as
$$F(u, v) = \sum_{y=0}^{(N-1)} \sum_{x=0}^{(M-1)} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$
 - $0 \leq u \leq M - 1$ and $0 \leq v \leq N - 1$
- ▶ The Inverse Discrete Fourier Transform of $F(u, v)$ is given as
$$f(x, y) = \sum_{v=0}^{(N-1)} \sum_{u=0}^{(M-1)} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$
 - $0 \leq x \leq M - 1$ and $0 \leq y \leq N - 1$
- ▶ As DFT is a complex valued function, $F(u, v) = R(u, v) + il(u, v)$
- ▶ $F(u, v) = r(u, v)e^{i\theta(u, v)}$ where
 - $r(u, v) = |F(u, v)|$
 - $\theta(u, v) = \tan^{(-1)}(I(u, v)/R(u, v))$

Transformed Domain Processing (cont.)





► Defn of Image addition

Let $f(x, y)$ and $g(x, y)$, where $0 \leq x \leq m - 1$ and $0 \leq y \leq n - 1$ be two images with values within 0 to $L - 1$.

The addition is defined as

- $(f + g)(x, y) = f(x, y) + g(x, y)$, where $0 \leq x \leq m - 1$ and $0 \leq y \leq n - 1$.
 - Transform $f + g$ to h with the range 0 to $(L-1)$, also with the property that $(f + g)(x, y) \leq (f + g)(x', y') \implies h(x, y) \leq h(x', y')$
- Note that the min and max values of f and g are 0 and $L - 1$ respectively, but the same is not true with $f + g$.
- It is required to transform the range to 0 to $L - 1$.



Application:

- ▶ CCD camera minimizes noise in the image captured using image averaging.
- ▶ It takes several images $f_i(x, y)$ $1 \leq i \leq k$, and finds
$$h(x, y) = \sum_{i=1}^k f_i(x, y)/k.$$
- ▶ If $f_i(x, y) = f(x, y) + N_i(x, y)$, where $N_i(x, y)$ is the noise added to actual image f , then
$$h(x, y) = f(x, y) + \sum_{i=1}^k N_i(x, y)/k.$$
- ▶ Since $N_i(x, y)$ will be positive and negative, $\sum_{i=1}^k N_i(x, y)$ will be a small value, and $\sum_{i=1}^k N_i(x, y)/k$ will be close to zero for large value of k .
- ▶ Hence $h(x, y)$ is a good approximation of $f(x, y)$.

Image subtraction

- ▶ **Defn:** The subtraction is defined as

$$(f - g)(x, y) = f(x, y) - g(x, y),$$

where $0 \leq x \leq m - 1$ and $0 \leq y \leq n - 1$, and transform the range of $f - g$ to integers from 0 to $L - 1$

- ▶ **Application:**

- Angiogram is an example where image difference is used . In Angiogram, heart is imaged using X-ray, say the image captured is $f_1(x, y)$, and another image is captured after injecting some contrast material into the blood stream.
- The contrast material will be absorbed more by the cholesterol content in the blood stream.
- The X-ray image of the heart after injecting contrast material will show the presence of cholesterol content.
- An another application of image subtraction is to detect moving objects.

- Moving objects in a video can be identified by taking difference of two consecutive frames.

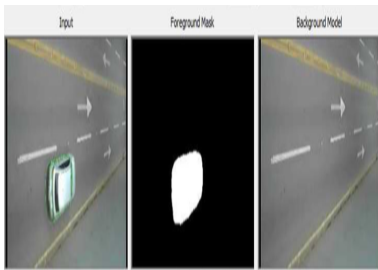
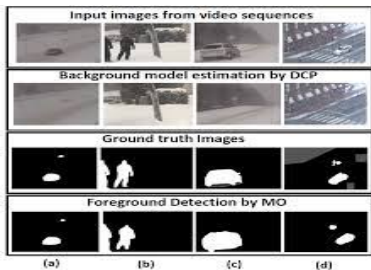


Image multiplication

- **Defn:** The image multiplication is defined as

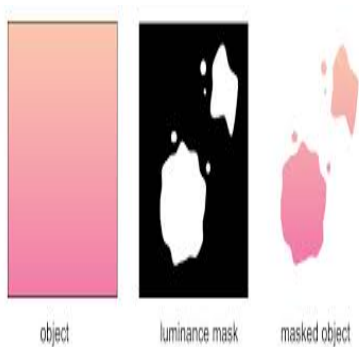
$$(fg)(x, y) = f(x, y)g(x, y),$$

where $0 \leq x \leq m - 1$ and $0 \leq y \leq n - 1$,

and transform the range of fg to integers from 0 to $L - 1$

- **Application:**

An application for image multiplication is found in Region of interest(ROI) detection. For instance, to extract image of a tooth from mouth image $f(x, y)$, define $g(x, y) = 1$ where (x, y) is possible locations of desired tooth; $g(x, y) = 0$ otherwise, and find $fg(x, y)$ which will give image of desired tooth image.



Common use of image multiplication is in masking also called as *region of interest* operations.



FIGURE 2.30 (a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

- ▶ **Defn:** The image division is defined as
$$(f/g)(x, y) = f(x, y)/g(x, y), \text{ if } g(x, y) > 0;$$
$$(f/g)(x, y) = f(x, y)/\epsilon \text{ otherwise,}$$
where $0 \leq x \leq m - 1$ and $0 \leq y \leq n - 1$, and transform the range of f/g to integers from 0 to $L - 1$
- ▶ Note that a small quantity ϵ is replacing 0 to make the division operation well defined.
- ▶ **Application:**
 - Usually the quality of image on screen projection by the LCD projector is degraded version of the actual image stored in computer.
 - The relationship between stored image $f(x, y)$ and the projected image $g(x, y)$ can be related as $g(x, y) = f(x, y)h(x, y)$, where $h(x, y)$ is the degrading function.
 - The original function $f(x, y)$ can be computed from $g(x, y)$ if we know $h(x, y)$ using image division $g/h(x, y)$.



- It is possible to estimate $h(x, y)$ from the known $f(x, y)$ and $g(x, y)$ for the fixed LCD Projector.

- Images are downloaded from internet sources

Thank You! :)